

Dishadharti — ML Technical Report

Climate screening for building proposals in Pune: data, methods, measurements, and sources. Every number below was read off a real run.

1. Problem and use cases

A municipal officer receives a development proposal and must decide: approve, condition, or reject. Today that decision has no climate evidence. This system scores the proposal on flood, heat, green cover, and embodied carbon (0–100, high = worse), explains the scores in plain language, and attaches redesign conditions whose effects are measured by re-scoring — not asserted.

Use case	Who	What the system does
Pre-approval screening	Municipal officer	Pin plot + two areas + bill of quantities; receive four scores, a verdict (approve / condition / reject), and ranked conditions.
Redesign guidance	Builder / architect	Each condition shows its measured score delta, so redesign targets the factor that moves most.
Tree compliance	Tree Officer	Plot-level tree ledger: what stays, what falls, age-equivalent compensation as a range.
Spoken explanation	Officer / public	Chatbot narrates the computed JSON. It reads numbers; it never computes or adjusts them.
Reuse in another city	Any municipal team	The engine holds no city knowledge: new config + cube + census + wards, same scorer.

2. System architecture

Two phases. Phase 1 runs once, offline: satellites and census files are baked into two artefacts — `epoch_cube.json` (864 one-km Pune cells: land cover, day/night surface temperature, elevation) and `tree_index.sqlite` (4,009,623 deduplicated census trees, spatially indexed). Phase 2 runs per proposal in ~10 seconds with no internet: point-in-polygon on 864 rows plus a tree-ledger walk, then deterministic arithmetic. This is a lookup table, not retrieval: no embeddings, no vector store. A committee can audit an exact number and challenge it.

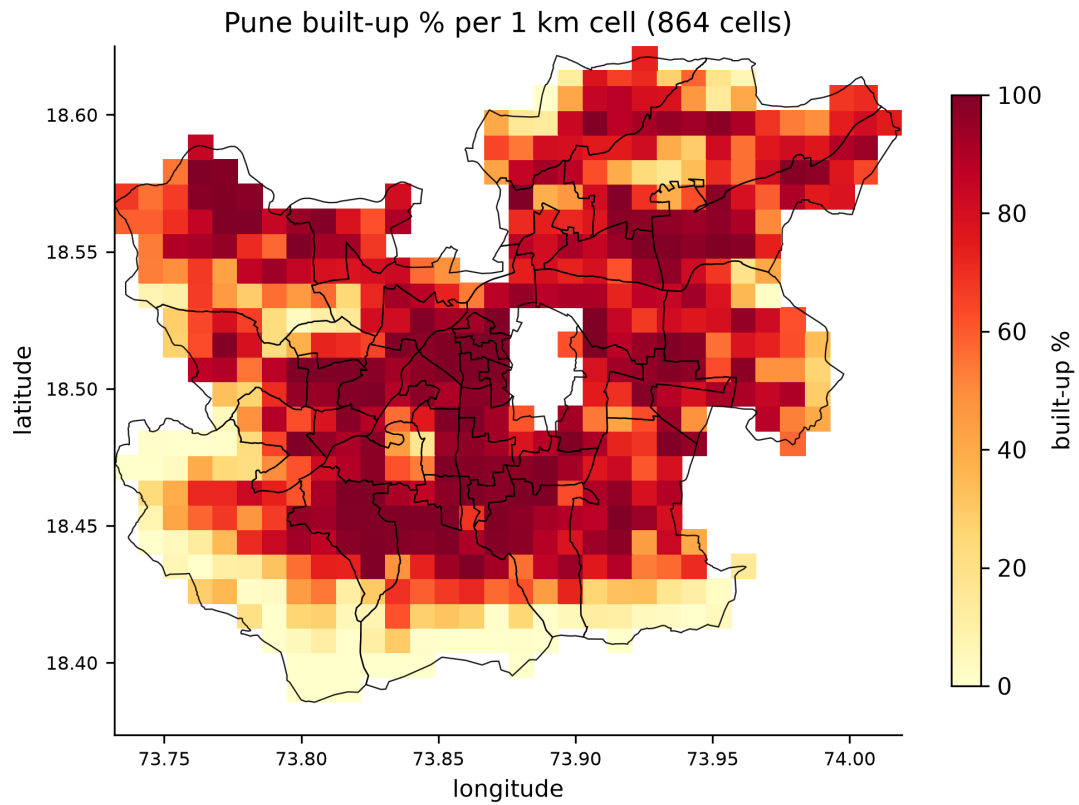


Figure 1. Built-up share per cell with ward boundaries. Every flood score in the system comes out of this partition.

3. Data built

Artefact	Content	Count / status
Epoch cube	2024 land cover + temperatures + elevation	864/864 cells complete
Tree index	Deduplicated 2019 census, spatial lookup	4,009,623 trees
Rainfall series	124 annual 1-day maxima, IMD grids	imd_pune_annual_max.csv
Emission table	38 materials, 102 aliases, India factors	materials.json
Species table	Census species data, full-index audit	species.json

The partition was the hard part. An early build silently deleted every water pixel, read vegetation from the wrong sensor at half its value, and smeared classifier uncertainty into phantom snow. All three were found by measurement and gated: the cube now refuses to build unless its nine land classes sum to one per cell. Water went 0 to 258 cells; vegetation median 6.3% to 30.2%.

4. Flood

Standard SCS curve-number hydrology (NRCS TR-55): area-weighted coefficients versus a vegetated baseline, excess runoff at the design storm. Drainage proximity can only lower the score; natural streams never count. Design storm 214 mm: Gumbel 100-year quantile over 124 IMD annual maxima (fit $p = 0.95$, interval 191–234). Across 41 wards: 14.5–53.3; built-up predicts it at $r = +0.92$, vegetation at $r = -0.95$.

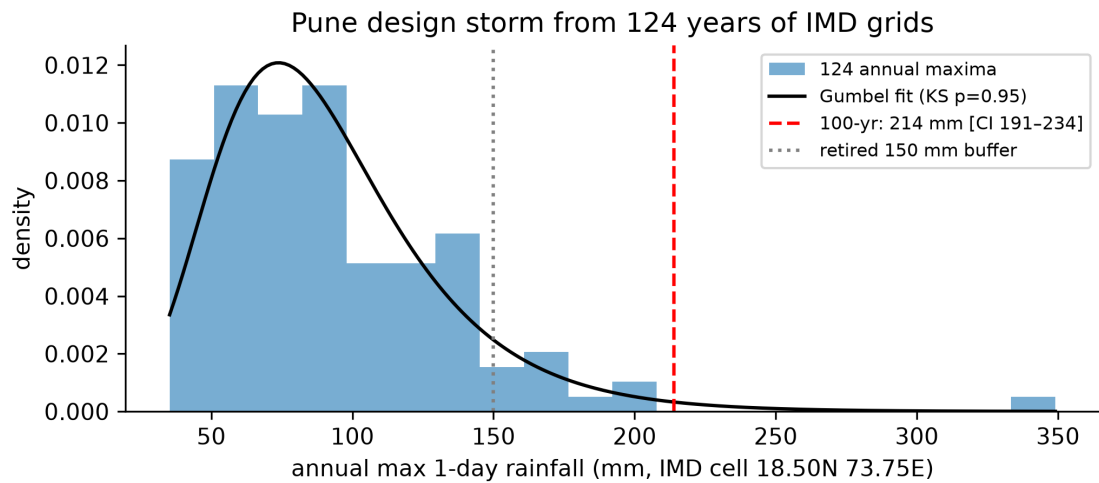


Figure 2. Gumbel fit to 124 IMD annual maxima. The retired 150 mm buffer sat between the 10- and 25-year storms.

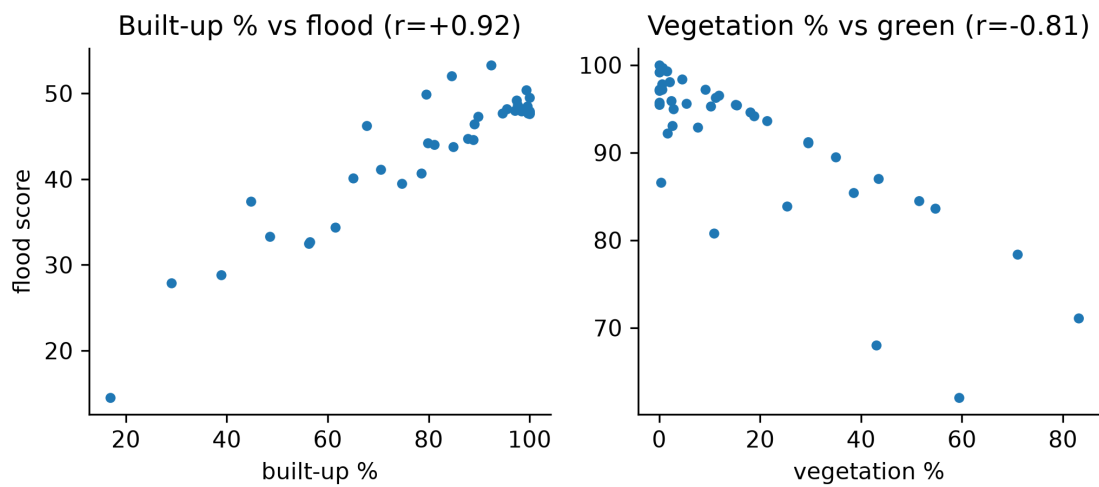


Figure 3. Flood rises with concrete, green cover falls with vegetation loss. Both point the right way.

5. Heat

Night surface temperature minus a rural vegetated ring, over a measured 5.0 C span. City mean +1.62 C above rural (Yale YCEO anchor sampled live at +1.139). Daytime was tested twice and retired: 11am contrast +0.11 C with seasonal sign flip; 1:30pm peak-hour contrast -0.14 C (dry rural soil out-heats shaded city — documented dryland effect). Reports label heat night-only and carry unscored afternoon context instead of a faked score. Spread 0.8–99.0, the widest of any factor.

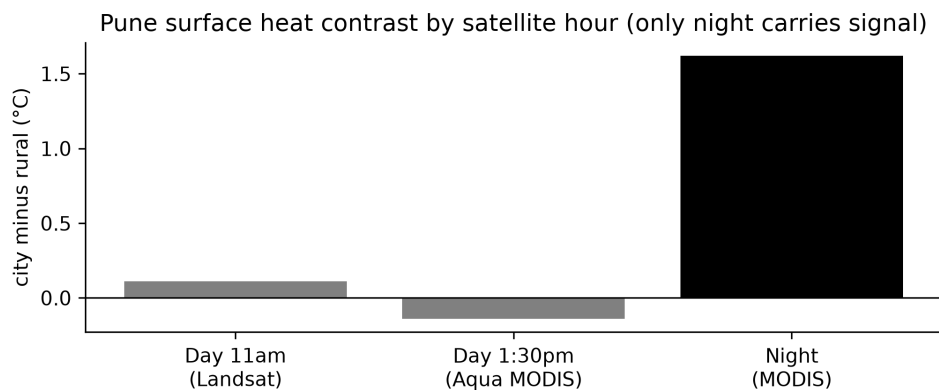


Figure 4. Only the night column carries signal.

6. Green cover and carbon

Green: census canopy (70%) + satellite ground vegetation (30%), stratified against double-counting; missing data returns unavailable, never perfect. Compensation is age-equivalent ranges under the 1975 Trees Act as amended 2021. Spread 62–100, $r = -0.81$. Carbon: bill of quantities times India factors (cement 0.91, steel 2.6, brick 0.39; aluminium 20.88), per square metre against a 1200 scale. Above 10% unknown mass the score is withheld. Flat across wards by design.

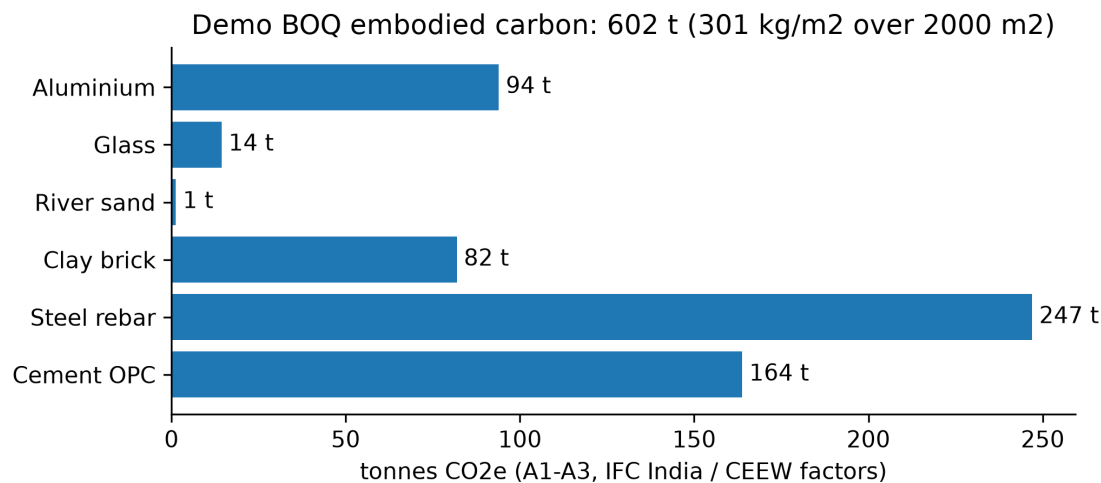


Figure 5. Demo bill: 602 tonnes, steel first. Every bar multiplies out by hand.

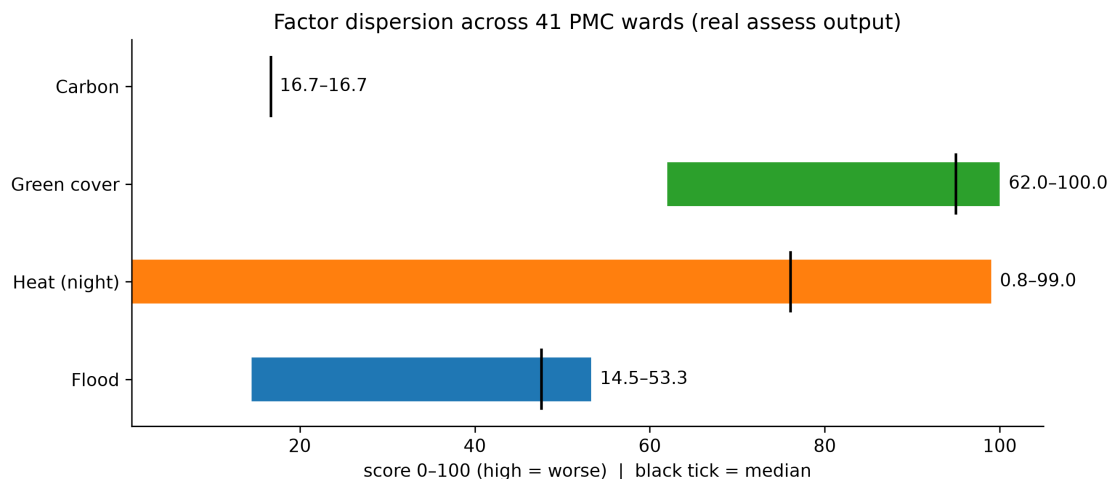


Figure 6. All four factors across 41 wards. Carbon is flat because materials do not vary by location.

7. What the system refuses

Missing data drops a factor out (weights redistribute) — never zero-filled. The narrator cannot alter numbers (prompt rules + tests; one live verdict-softening caught and locked). Unsourced numbers stay labelled (carbon bands, soil group). No neural scorer: one candidate heat model was trained, failed its own bars, and ships nothing.

8. Verification ledger

Claim	Evidence
Flood tracks built cover	$r = +0.92$, $n = 41$ real runs
Flood tracks vegetation	$r = -0.95$
Green tracks vegetation	$r = -0.81$

Claim	Evidence
Night heat has contrast	city +1.62; $r = +0.55 / -0.54$
YCEO anchor	+1.139 sampled live
Storm 214 mm	Gumbel KS $p = 0.95$, CI 191–234
Cement / steel / brick	IFC India Table 16
IGBC / high-rise anchors	700 / 454, primary sources
Census total	4,009,623 indexed of 4,009,624 unique
Suites	48/48, 8/8, e2e, narrative 22/22, engine 23/23

9. Sources

Datasets

Dataset	Source
PMC Tree Census 2019 (17 parts)	data.opencity.in, CKAN f00d83b8-c70f-4fff-9ac7-9a6ab4255edf
IMD 0.25-degree daily rainfall 1901–2024	imd pune.gov.in/cmpg/Griddata (Pai et al. 2014, MAUSAM)
Landsat 8/9 C2L2 surface temperature	USGS; GEE LANDSAT/LC08/C02/T1_L2, LC09/C02/T1_L2
MODIS day/night LST (Terra + Aqua)	NASA; GEE MODIS/061/MOD11A1, MYD11A1 (overpasses 10:30 / 1:30)
Dynamic World land cover	Google; GOOGLE/DYNAMICWORLD/V1
ECOSTRESS night LST	NASA/JPL; NASA/ECOSTRESS/L2T_LSTE/V2
Yale YCEO urban heat	YALE/YCEO/UHI/UHI_yearly_averaged/v4
GEDI canopy height	NASA; LARSE/GEDI/GEDI02_A_002_MONTHLY
IFC India materials database	IFC/EU Eco-cities 2017, Table 16 GWP
PMC ward boundaries	pmc_wards_2025.kml (41 wards)

Papers and standards

Reference	Used for
USDA SCS TR-55 (1986)	Curve numbers, runoff equation
Vivekanandan (2022), JWRE	Gumbel/EV1 choice for Pune
Das et al. (2022), MAUSAM	Gumbel IDF precedent, India
Chow / Maidment / Mays (1988)	Annual-maximum method
Stewart & Oke (2012)	Local Climate Zones language
Karadumpa et al. (2024)	Indian cement plant factors
CEEW (steel; aluminium)	Indian industry factors
Akshatha et al. (2025)	High-rise mean 454
IGBC Net Zero Carbon rating	700 kgCO2e/m2 anchor
WRI India Pune tree-cover note	620 ha loss; census caveats
Maharashtra Trees Act 1975 + 2021 amendment	Felling permission; age-equivalent compensation

10. Run and reuse

Assess: python -m ml_pipeline.cli assess --ward 12 --built-up 3000 --plot 8000 --materials '['...']' --role municipal_authority --json-out /tmp/out.json. Narrate: python ml_pipeline/report/narrative.py /tmp/out.json (GEMINI_API_KEY in .env). Any other city: engine/ takes a config, a cube, a census index, and ward polygons — the scorer holds no city knowledge.

Known limits. Flood and heat score a 1 km square, not the plot. Daytime surface heat carries no signal here. Carbon bands are house bands with sourced anchors. A plausible-but-wrong input inside 0–100 is undetectable without a second source. No trained model exists here, deliberately.